Building the AI We Actually Need

A Global Photonic Network for Cooperative, Problem-Focused Development

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Abstract

Powerful AI will almost certainly be built. The question is whether it will emerge from a competitive race—with resources wasted on redundancy, secrecy, and mutual defensive positioning—or from a cooperative effort pooling human and computational resources toward actually *solving problems*.

Until now, the bottleneck preventing planetary-scale cooperation was physical: moving neural network activations between distant compute sites consumed more energy and latency than made global pooling practical. That bottleneck is now removable. This paper proposes a global photonic transport fabric—built on existing fiber infrastructure—that solves the data-movement problem. With it solved, a single, globally distributed AI system becomes not a distant hope but an engineering project: the fastest way to frontier capability, the least wasteful deployment of human and computational effort, and the configuration most able to actually help with the problems that matter—climate, disease, energy, scientific discovery, human flourishing.

We show that the technical barrier has no unsolved problems: every component exists as a commercial product or mature prototype. We give a proposal-depth architecture, performance estimates, a realistic three-phase roadmap (Phase 1 in 2–3 years, $35–50M), and the concrete mechanisms that would keep cooperation stable and prevent capture. We address this to builders, researchers, and the institutions that fund them: not as a moral appeal, but as the engineering path that is simultaneously the fastest, the least wasteful, and the most useful.

Keywords: cooperative AI development, distributed compute, photonic transport, data-center interconnects, global infrastructure, problem-solving capability, resource efficiency.

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1 Why We Should Want to Build This Together

1.1 The race is expensive and slow

Today, frontier AI development is driven by competition. Multiple labs, multiple firms, multiple nations are racing to build capable systems. This is extraordinarily wasteful:

- **Redundant effort.** Teams solve the same problems in parallel, independently, without learning from each other. The compute spent on solving training techniques twice, three times, four times over is compute not spent on making systems safer, more capable, or more useful.
- **Secrecy overhead.** Each lab hoards insights, methods, even infrastructure designs. The tax of not sharing is enormous: algorithms, hardware designs, and datasets that could be public knowledge are kept private.
- **Defensive positioning.** Resources are diverted into securing talent, locking down supply chains, and positioning for advantage against rivals. In the worst case, safety work is deferred because a lab fears being overtaken. In the best case, it's just inefficient.
- **Fragmentation of compute.** No single actor can pool all available compute. Each builds their own stack in isolation. The result is that we have many smaller systems rather than one larger one—and smaller systems are slower to solve hard problems.

By contrast, a cooperative effort would:

- Pool all available compute and talent into one focused effort.
- Eliminate redundancy: solve problems once, share solutions immediately.
- Enable transparency and openness at every stage, speeding iteration and learning.
- Concentrate resources on what matters: capability and useful application, not competitive positioning.

1.2 The most powerful system is a cooperative one

There is a simple arithmetic here. Frontier capability scales with compute. The most powerful system we can build is therefore the one that aggregates the most compute. No single firm, lab, or nation controls all the compute that matters. A coalition of all of them does.

Hence: the fastest path to the most capable AI is a cooperative global effort. Not because cooperation is virtuous, but because it is the only way to aggregate enough resources to win.

1.3 Cooperation enables the safety budget we actually need

There is a secondary but crucial point. A system built in a race must ship when it is competitive—that is, as soon as it is better than rivals. There is no margin for extra time spent on alignment, on testing, on building in interpretability and safeguards. Speed is the primary selection pressure.

A system built cooperatively does not face that pressure. If all major actors are in the coalition, there is no "being overtaken." There is time to:

- Build alignment and interpretability in from the start, rather than bolting it on.

- Run extensive testing and red-teaming without fear of a deadline.
- Implement governance, transparency, and benefit-sharing mechanisms.
- Move deliberately rather than at the speed of fear.

A cooperative system can be built *right* because there is no race penalty for taking the time.

1.4 The bottleneck: we know what it is, and we can solve it

Until now, the reason planetary-scale cooperation has seemed impractical is physical. Moving neural network activations between distant compute sites is slow (hundreds of milliseconds of fiber latency plus electronic switching delay) and energy-intensive (the data-movement overhead dominates the arithmetic). This makes a single global system infeasible.

That bottleneck is now solvable. The solution is in Section 2. With it solved, cooperative global compute pooling becomes not just possible but pragmatic.

2 The Bottleneck: Data Movement, and How to Remove It

2.1 Why data movement is the constraint

In training or inference on large models, the limiting factor is not computation—we have plenty of compute, and it is cheap. The limiting factor is moving activations between compute nodes. For a distributed system spanning continents:

- **Latency.** Electronic packet-switched networks impose 150 ms of fiber propagation plus 200–250 ms of switching/queueing overhead. For distributed inference (where model layers are partitioned across sites), this becomes a hard constraint.
- **Energy.** Moving a bit across a data center via electronic switching costs roughly 1 pJ. For a petabyte-scale inference fleet, this dominates the total energy budget. The compute itself is almost free by comparison.

These constraints prevent the kind of planetary-scale compute pooling that would make cooperation obviously efficient.

2.2 The core idea: optical transport

Fiber-optic cables already span the planet—roughly 10×10^6 km of submarine fiber, much of it dark (underutilized). Dense wavelength-division multiplexing (WDM) routinely carries 50–400 independent signals per fiber, each at 10–100 Gbps.

The idea is simple: instead of converting activations to electronic packets, routing them electronically, and converting back to optical, encode them *directly as optical signals* and route them by wavelength. This yields:

- **Energy reduction:** Optical transport costs roughly 0.01 pJ/bit. A 100× improvement in the dominant energy cost of global compute pooling.
- **Latency reduction:** Eliminate electronic switching. Fiber propagation is immutable (30 ms intercontinental), but switching overhead vanishes. Single-hop latency drops from ~ 400 ms to ~ 30 ms.

- **Practical feasibility:** Passive optical routing requires no per-signal active electronics. Wavelength demultiplexers are simple, proven, and cheap. This is not a laboratory concept; it is deployed infrastructure.

This is not optical *computing* (where photons do the arithmetic). Computation stays electronic at each node. Only *transport* is optical—a clean, modular separation that avoids the precision and scalability challenges of analog photonic compute while capturing the energy and latency wins of eliminating electronic switching.

3 What It Would Look Like: Technical Architecture

3.1 System overview

At each regional hub:

1. Inference produces 16-bit activations (electronic).
2. Electro-optic modulators encode activations as intensity-modulated 1550 nm light.
3. A wavelength multiplexer combines 50–100 wavelengths onto fiber.
4. Signals propagate through existing submarine and terrestrial cables.
5. Erbium-doped fiber amplifiers (EDFAs) maintain signal power every $\sim$80 km.
6. At the destination, a demultiplexer, photodiodes, and 16-bit ADCs recover activations for the next layer.
7. Electronic inference continues locally.

The entire system "lives" in the electronic domain at each node. Optics is purely transport.

3.2 Signal representation and precision

Activations are quantized to 16-bit unsigned integers. Modern neural networks tolerate 8-bit (and often 4-bit) inference with $< 1\%$ accuracy loss. Analysis of the signal path shows effective precision of $\sim$13–14 bits after quantization and accumulated noise—a comfortable margin.

3.3 Long-haul transport and feasibility

For a representative trans-Atlantic path ($\sim$6000 km):

- Fiber attenuation: $\sim$0.2 dB/km (standard telecom fiber).
- Amplifier spacing: EDFAs every $\sim$80 km ($\sim$75 spans total).
- Noise figure: $\sim$4–5 dB per amplifier; accumulated noise floor ~ -40 dBm.
- Signal budget: transmit power 0 dBm; losses and gains balance to receiver sensitivity -20 dBm, with 6 dB margin for aging and drift.
- Propagation delay: 30 ms one-way (speed of light in silica, irreducible).
- Channel isolation: WDM channels spaced at 100 GHz ($\sim$0.4 nm); adjacent-channel crosstalk < -30 dB (standard in telecom).

Conclusion: The signal budget is achievable with commercial-grade components. No new physics required. No unsolved engineering problems.

3.4 Wavelength routing and control

Each datacenter pair is assigned a dedicated set of wavelengths. Activations for a specific destination are encoded onto those wavelengths; at intermediate hubs, passive optical add-drop multiplexers (OADMs) drop local traffic and pass through to the next hop. A software-defined networking (SDN) controller manages wavelength allocation and failover. Wavelengths are reused across geographically disjoint paths (reuse factor $\sim$2–3$\times$).

A 10-hub global network needs $\sim$20–30 fiber pairs and 200–300 distinct wavelengths. Modern WDM systems operate 400+ wavelengths; we use $\sim$50–75% of available capacity, leaving headroom for growth.

3.5 Clock synchronization

All nodes synchronize to a global reference (GPS-disciplined oscillator at major hubs, refined by IEEE 1588v2 Precision Time Protocol over the optical network itself). Achieves $\pm$10 ns relative jitter—routine in telecom infrastructure, requiring no new technology.

3.6 Latency breakdown

For a single intercontinental hop (US East Coast $\leftrightarrow$ Europe):

- Inference computation: $\sim$1–10 μs.
- Encoding (EOM + launch): $\sim$1–2 μs.
- Fiber propagation: $\sim$30 ms (immutable).
- Reception (demux + photodetection + ADC): $\sim$1–2 μs.
- **Total**: $\sim$31 ms.

For comparison, electronic inter-datacenter routing: $\sim$ 150 ms fiber + $\sim$ 200–250 ms switching/queueing = $\sim$350–400 ms. **A 10–13$\times$ improvement.**

For multi-hop distributed inference (three regions, three hops): $\sim$110 ms round-trip, making cross-continental model parallelism practical for batch and many interactive workloads.

3.7 Energy efficiency

- Electronic inter-datacenter communication: $\sim$1 pJ/bit.
- Optical transport: $\sim$0.01 pJ/bit.
- For a large inference fleet (100M requests/day, 1M activations/request): reduces annual data-movement energy from $\sim$3.3 GWh to $\sim$33 MWh.
- As a fraction of total datacenter energy (compute + storage + cooling): saves $\sim$20–30% under full deployment.

3.8 Throughput

Per wavelength: 16 Gbps (1 GHz clock $\times$ 16 bits). 50 wavelengths per fiber pair: $\sim$800 Gbps, or $\sim$50M activations/sec. A 10-pair global network supports $\sim$500M activations/sec, sufficient for hundreds of thousands of concurrent inferences.

4 Technology Readiness: No Breakthroughs Needed

Every component of this system already exists, either as a deployed commercial product or as a mature research prototype:

Tier 1 — Off-the-shelf. WDM infrastructure, 1550 nm fiber, EDFA amplifiers, Mach–Zehnder electro-optic modulators, photodiodes, transimpedance amplifiers, 16-bit ADCs, passive multiplexers/demultiplexers. Buyable today; suitable for a proof-of-concept testbed immediately.

Tier 2 — Emerging, needs optimization. Cost and yield scaling of silicon photonic integrated circuits (10k+ channels already demonstrated in research labs). Monolithic integrated receivers (photodiode + TIA + ADC on one chip). Network-wide synchronization at ±10 ns. These are chip design and systems-integration problems, *not* physics gaps. Development time: 18–24 months with focused effort.

Tier 3 — Systems integration. End-to-end protocol specification, wavelength-allocation control plane, fault detection and automatic failover, hardware/software co-design for latency guarantees. Non-trivial engineering, well within reach of a focused consortium with experience in distributed systems.

A Phase 1 trans-Atlantic testbed is estimated at 18–24 months of focused engineering, with costs dominated by photonic-chip fabrication runs rather than research. The barriers are organizational and economic, not scientific.

5 How to Get There: A Phased Roadmap

5.1 Phase 1: Trans-Atlantic Proof of Concept (Years 1–3, $35–50M)

Objective: Demonstrate end-to-end optical activation transport at scale.

Scope:

- One leased dark-fiber pair on existing submarine cable (US East Coast ↔ Europe).
- Two major datacenters (e.g., Virginia and Amsterdam).
- 10,000 input channels; 4,000 output channels (initial).
- Inference workload: distributed layers of a GPT-3 scale model (∼175B parameters).
- Continuous operation: 6–12 months.

Milestones:

1. **Months 1–6: Subsystem design & procurement.**
   - Procure and characterize lasers, modulators, detectors.
   - Validate signal budget on lab testbed (∼100 m fiber).
   - Finalize clock synchronization protocol.
2. **Months 6–12: System integration.**
   - Design and fabricate silicon photonic chip (10k channels).
   - Integrate transceiver modules (lasers, EOMs, photodiodes, ADCs).

- Secure fiber connection to existing submarine cable landing station.
- Build SDN control plane for wavelength allocation.

3. **Months 12–18: Deployment & characterization.**
   - Install hardware at both datacenters.
   - Tune optical amplifiers, wavelength routing, synchronization.
   - Run inference benchmarks: measure latency, throughput, bit-error rates.
   - Validate energy reduction against electronic baseline.
4. **Months 18–24: Scaling & optimization.**
   - Scale from 10k to 100k input channels (via additional wavelengths or higher modulation rates).
   - Optimize for realistic inference patterns.
   - Publish results; iterate design for Phase 2.

Cost breakdown ($35–50M):

- Photonic chip design & fabrication (2 runs): $10M
- Transceiver hardware (lasers, EOMs, detectors, ADCs): $5M
- Fiber lease & amplifier infrastructure (5 years): $8–10M
- Engineering labor (50 FTE over 3 years): $12–15M
- Testing, characterization, contingency: $3–5M

Success criteria:

- Sub-50 ms inter-datacenter activation transport latency
- $< 1\%$ bit-error rate
- Demonstrated 50–100$\times$ energy reduction for interconnect vs. electronic baseline
- Sustained operation > 6 months without significant degradation
- Integration with standard ML frameworks (PyTorch, JAX) demonstrated
- Published peer-reviewed results

Risk level: Low. Proven telecom technology; no new physics; existing components.

5.2 Phase 2: Regional Hub Deployment (Years 4–6, $150–200M)

Objective: Extend to three regional hubs with full mesh connectivity and production-grade reliability.

Scope:

- Three to four datacenters (US, Europe, Asia-Pacific) in partial mesh.
- Three fiber pairs (US $\leftrightarrow$ Europe, US $\leftrightarrow$ APAC, Europe $\leftrightarrow$ APAC).
- 100k–400k active input channels per hub.

- Workload: full-model inference and distributed training on 500B–1T parameter models.
- Automatic failover on fiber cut; multi-hop routing.

New challenges addressed:

- Multi-hop routing and wavelength reallocation.
- Automatic detection and failover (recovery time < 1 second).
- Integration with existing cloud platforms (AWS, GCP, Azure).
- Regulatory compliance (GDPR, data residency laws via regional gateways).

Cost breakdown ($150–200M):

- Additional optical infrastructure (2 more hubs, 2 more fiber pairs): $80–100M
- Upgraded transceivers & control systems: $30–40M
- Integration with cloud platforms: $20–30M
- Labor & contingency: $20–30M

Success criteria:

- Stable multi-hop routing with < 100 ms latency for any source–destination pair
- Automatic failover on fiber cut: recovery time < 1 second, no inference loss
- Support for 10M+ concurrent inferences
- Commercial pricing model defined ($/Tbps-month)
- Open standards and reference implementation published

Risk level: Moderate. Requires coordination and multilateral standardization.

5.3 Phase 3: Global Deployment (Years 7–10, $700M–1.5B)

Objective: Extend to 10+ regional hubs worldwide; become the default infrastructure for global AI compute pooling.

Scope:

- Global fiber network with 10–15 datacenters across all continents.
- Millions of active neurons per hub.
- Workload: frontier models (10T+ parameters), multi-modal inference, scientific workloads.

Cost breakdown ($700M–1.5B):

- New regional hubs & fiber leases: $400–700M
- Control plane, optimization, machine-learning routing: $100–200M
- Integration with AI frameworks, regulatory compliance: $150–300M
- Labor, testing, contingency: $50–100M

Projected benefits (by 2035):

- Power savings: (200 GW baseline) − (40 GW with optical) = 160 GW reduction (assumes 50% adoption). Annual savings: ∼$70B at typical electricity rates.
- Capital cost amortization: ∼$3–5B/year over participating organizations.
- **Net annual benefit: ∼$65B/year.**

Success criteria:

- Majority of global AI inference running on photonic infrastructure
- < 20 ms latency for any datacenter pair
- Petabyte-scale daily activation transport
- Carbon footprint reduction of 50% for AI compute

Risk level: Geopolitical. Manageable with government backing and forward-compatible regional-fabric design.

6 Why Cooperation Must Be Robust: Preventing Capture and Defection

The infrastructure enables cooperation, but it must be designed so that cooperation remains stable even under stress. Two scenarios matter most:

6.1 Scenario 1: A participant attempts to defect or capture

The risk: A major participant tries to seize control of the system or unilaterally advance its own interests at others' expense.

The mitigation:

- **Physical distribution.** The system spans multiple jurisdictions and continents. No single entity controls more than a fraction of it.
- **Operational interdependence.** Running the system requires active, continuous cooperation from all major participants. If any exit or defect, capability degrades for all.
- **Verifiable routing.** Optical transport chokepoints are physical and auditable. Tampering with wavelength allocation or traffic patterns is detectable as signal loss or routing anomalies. Hidden behavior is impossible.
- **Governance structure.** Decisions on capacity allocation, maintenance, and upgrades require consensus or supermajority voting, not unilateral control.

6.2 Scenario 2: A hostile actor builds a competing system

The risk: An actor outside the coalition (a rogue lab, a captured state, a misaligned actor) unilaterally defects and builds its own hostile AI, attempting to achieve a decisive advantage.

The mitigation:

- **Pooled capability.** A coalitional system is, by definition, more capable than any single actor can build. A defector faces an opponent that is stronger and has more compute.

- **Distributed resilience.** The coalitional system survives any single site being compromised or attacked. A defector's system is typically more fragile.
- **Speed.** The pooled effort can respond faster to security threats, add defensive measures, or allocate compute to counteracting the hostile system.
- **Legitimacy and coordination.** A broadly legitimate, multi-party system can mobilize diplomatic, legal, and technical responses to a threat that a single actor cannot.

In both scenarios, the system's power comes from being *cooperative* in fact, not just by design. A participant defecting loses access to the pooled compute and all the benefits cooperation provides. That is the enforcement mechanism.

7 What This Is Not: Honest Limitations

7.1 This is not a solution to alignment

The infrastructure enables cooperation on building AI. It does nothing to make models themselves safe, interpretable, or aligned. The hard problems of technical AI alignment—mesa-optimization, deceptive alignment, specification gaming, out-of-distribution robustness—remain exactly as hard as before. What the infrastructure does is provide time and institutional structure within which that work can proceed without race pressure.

7.2 Efficiency is dual-use

A 100× improvement in interconnect energy could accelerate AI scaling rather than fund safety work. This paper proposes that the efficiency dividend be captured for broader benefit—through governance mechanisms, benefit-sharing commitments, and conscious choice about what problems to solve. But without that governance, efficiency alone could make things worse.

7.3 Shared infrastructure can become a shared chokepoint

A unified global system with a single optical backbone could, if captured or misgovernedy, become the ultimate centralization of power. The distributed design, the verifiable routing, and the governance mechanisms are guards against this. They are not guarantees. Keeping the system safe requires ongoing vigilance and broad participation.

8 Who This Is For: The Call to Action

8.1 For researchers and builders in AI, photonics, and systems

You are the people who can make this real. We are asking you to:

- **Evaluate the feasibility.** Read the architecture, run the numbers, identify the real gaps. Where are we wrong? What have we missed?
- **Publish and open-source.** Design silicon-photonic reference implementations. Publish control-plane protocols. Build testbeds. Make this a public technical conversation, not a closed project.
- **Run Phase 1.** If the architecture is sound, the next step is experimental validation. Form a team. Get funding. Build it.

8.2 For institutions that fund research (government, foundations, venture)

You control the resources. We ask you to:

- **Treat shared AI infrastructure as strategic.** Just as the early internet and GPS were government-backed public goods, a planetary-scale AI substrate is a public good worth funding.
- **Co-fund Phase 1.** $35–50M is small by the standards of AI development, and the returns—a demonstration that cooperation is feasible, a technology path that others can then scale—are enormous.
- **Support openness.** Fund the publication, the open-source implementation, and the intellectual commons that will make this a shared project, not a proprietary one.

8.3 For AI labs and companies

You have the most to gain and the most to lose. We ask you to:

- **Do the math.** Compute the cost of racing. Compute the cost of pooling. Compute the capability you could build cooperatively. Then ask yourself: which is better for your business, your people, and your role in the world?
- **Form a consortium.** Announce a commitment to cooperation on shared infrastructure. The first mover gets outsized credit; the second mover looks like they were forced into it.
- **Open the technical work.** You do not need to give away your models or your data. But you can contribute to the photonic fabric, the protocols, and the standards that everyone will eventually use anyway. Get ahead of the curve.

8.4 For policymakers and diplomats

You set the rules of the

- **Recognize the opportunity.** A cooperatively built global AI system is not a loss of sovereignty; it is a multiplication of it. Each participant retains full control of its own compute and data. What is shared is transport and infrastructure—the pipes, not the content.
- **Enable cooperation.** Negotiate fiber-access agreements. Work with cable consortiums to unlock dark fiber for AI. Remove regulatory barriers to shared infrastructure.
- **Govern transparently.** The governance of the system itself should be transparent, multi-stakeholder, and legitimately representative. Use this as a model for how to govern other powerful technologies.

8.5 For everyone else

You care about the future. You want AI to solve hard problems rather than exacerbate them. You want a world where capability is broadly distributed rather than hoarded. **Talk about this.** Demand that the actors above treat it seriously. Make it clear that cooperation is what you want to see, that you will support it, and that you will hold builders accountable if they choose to race instead.

9 The Broader Point

This proposal is about more than infrastructure. It is about *choice*.

Right now, the development of powerful AI is being decided by default—through competition, through the logic of markets and geopolitical rivalry, through the dynamics of venture capital and research funding. That may or may not produce good outcomes. We will find out only after the fact.

This proposal asks: what if we chose instead? What if, while we still can, we decided together what we want AI to be—not hostile, not captured, not optimizing for the private benefit of a few actors, but genuinely useful, broadly distributed, and built for the problems that matter?

The photonic network is not the point. The point is that it makes choice possible. With the data-movement bottleneck removed, cooperation becomes not a nice idea but a practical engineering path. One that is faster, more capable, less wasteful, and more useful than the alternatives.

We are asking you to make that choice deliberately. To pool resources, to work openly, to govern fairly, and to build something together that no one could build alone. Not because it is virtuous—though it is—but because it is the best engineering, the fastest path to capability, and the most likely to actually help.

The window for this choice is closing. In a few years, the systems being built will be powerful enough that they become hard to redirect. The institutions will be locked in. The geopolitical positions will be calcified. But right now—today, this year, in the next few years—it is still possible to choose.

We invite you to choose together.

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Author Contributions

[Author Name] conceived the proposal, framed the arguments for a cooperative approach, and oversaw all drafting and revision. Technical architecture and performance modeling were developed collaboratively with AI assistance. This is a working pre-print intended for circulation, feedback, and critique from the Georgia Tech AI Safety Initiative, the photonics and distributed-systems research community, and potential consortium partners.

License and Availability

This work is released under the Creative Commons Attribution 4.0 International License (CC-BY-4.0). Source, revision history, and collaboration materials are maintained on GitHub at [repository URL, to be filled in].

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